



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2023-24(Even Semester)

| VISION                                                                                       | MISSION                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To emerge as a learning hub and center of excellence in the domain of Information Technology | <p>[M1] To impart quality technical education through effective teaching learning process.</p> <p>[M2] To provide a platform to address societal issues as well as challenges faced by IT industries.</p> <p>[M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT.</p> <p>[M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.</p> |

### Class Test (CT)-2

Semester: IV

Programme: B. Tech Information Technology

Course: Database Management System

Time: 1 Hr 60 min

Max Marks: 30 Marks

#### Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

| BIT32402 | Course Outcome                                                                                 |
|----------|------------------------------------------------------------------------------------------------|
| CO1      | Understand the fundamental of database system and DBMS concepts.                               |
| CO2      | Identify the concept of conceptual data modeling using the E/R model.                          |
| CO3      | Illustrate the concept of embedded SQL and its role in application development.                |
| CO4      | Analyze the process of database normalization and the steps for achieving 1NF,2NF,3NF,and BCNF |
| CO5      | Compare Various indexing techniques like hash-based indexing,dynamic hashing,and B+trees.      |

| Group A |                                                                                                                                                            |     |     |       |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|
| Que. 1  | Answer the following Question                                                                                                                              | CO  | BTL | Marks |
| a.      | Discuss the Purpose of primary key and foreign key constraint.                                                                                             | CO3 | 2   | 2     |
| b.      | Illustrate Multivalued Dependency.                                                                                                                         | CO4 | 2   | 2     |
| c.      | Classify the file organization any name any two types.                                                                                                     | CO5 | 2   | 2     |
| Group B |                                                                                                                                                            |     |     |       |
| Que. 2  | Answer the following Question (Solve any Two)                                                                                                              | CO  | BTL | Marks |
| a.      | Write SQL query using GROUP BY and HAVING to<br>a) Find the department wise average salary<br>b) Display the department name having more than 10 employee. | CO3 | 2   | 4     |

|        |                                                                                                                                               |     |     |       |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|
| b.     | Explain various aggregate function with suitable example.                                                                                     | CO3 | 2   | 4     |
| c.     | Enlist the properties of transaction.                                                                                                         | CO3 | 2   | 4     |
| Que 3. | Answer the following Question (Solve any Two)                                                                                                 | CO  | BTL | Marks |
| a.     | Interpret functional dependency and describe its various types.                                                                               | CO4 | 2   | 4     |
| b.     | Classify the different types of normalization and demonstrate how BCNF differs from 3NF with a suitable examples.                             | CO4 | 3   | 4     |
| c.     | Illustrate join dependencies in the context of the Fifth Normal (5NF) with example.                                                           | CO4 | 3   | 4     |
| Que 4. | Answer the following Question (Solve any Two)                                                                                                 | CO  | BTL | Marks |
| a.     | Illustrate deadlock detection and prevention techniques.                                                                                      | CO5 | 2   | 4     |
| b.     | Implement a well – labeled diagram to represent the ACID properties of a transaction and analyze how they contribute to database reliability. | CO5 | 3   | 4     |
| c.     | Examine the importance of serializability in ensuring consistency in transactions with a suitable example.                                    | CO5 | 4   | 4     |

**Bloom's Taxonomy Level (BTL)-** Remember , Understand , Apply, Analyze , Evaluate , Create